

Sarah Jenkins

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EDUCATION

University of Texas at Austin

B.S. in Electrical and Computer Engineering

Austin, TX

Graduation: May 2026

TECHNICAL SKILLS

Hardware/HDLs: SystemVerilog, VHDL, Assembly (ARM), MATLAB, Verilog

Languages: C, C++, Embedded Python, Shell Scripting

Tools/Labs: Xilinx Vivado, ModelSim, Oscilloscopes, Logic Analyzers, FreeRTOS, Git

Focus Areas: RTL Design, FPGA Synthesis, Computer Architecture, Embedded Firmware

TECHNICAL PROJECTS

8-Bit Pipelined CPU Architecture Core

Fall 2025

Lead Hardware Designer

- Implemented a fully functional pipelined RISC processor model from architectural scratch using **SystemVerilog**.
- Resolved instruction hazard behaviors via explicit forward bypass logic gates and stall condition monitoring components.
- Validated architectural structural fidelity inside ModelSim running intensive mock binary assembly tests.

Real-Time Sensor Ingestion Firmware

Spring 2025

Embedded Systems Class Project

- Authored precise multi-threaded data capture routine on an ARM Cortex microchip running a **FreeRTOS** kernel.
- Wrote custom low-level device driver modules in **C** handling peripheral communication over SPI and I2C buses.
- Managed low-power execution parameters reducing battery consumption characteristics during idle loop execution timeouts.

EXPERIENCE

Undergraduate Engineering Tutor

September 2024 – May 2025

UT Austin ECE Department

Austin, TX

- Provided rigorous structural troubleshooting assistance for 60+ undergraduate peers building basic state machines.
- Evaluated introductory lab scripts grading execution accuracy on digital design boards.